Electrostatic Linear Servo Motor with Built-in Position Sensor for Vacuum Environment

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Abstract— **A high-power electrostatic linear servo motor stage for vacuum environment has been developed. The stage is equipped with integrated built-in position sensor that uses the motor's electrode for position sensing. This stage is characterized by compact structure and low magnetic field emission. It can work within sufficiently small space in a vacuum chamber without disturbing surrounding electro-magnetic field. Therefore this motor stage could be highly compatible with many vacuum applications that utilize electron beam.**

In this paper, basic structure of the motor is described. The motor is sized 145mm in length, 125mm in width, and 40mm in height. The stage is equipped with two thin plastic films, which contain three-phase electrodes aligned with an interval of 200μm. Those two films are attached to slider and stator, respectively, to generate electrostatic thrust force. The slider is supported by linear cross-roller guide. The stroke is 40mm. The estimated thrust force in 10^{-3}Pa is 1.2N with an applied voltage of $1.6kV_{0-p}$. We have developed built-in position sensor for positioning servo control. Detail of built-in position sensor and results of positioning control are reported.

Index Terms—**Electrostatic motor, Actuator, Vacuum, Positioning, Electron beam lithography**

I. Introduction

Recently, progress in nano science and further miniaturization of IC patterns require specially conditioned vacuum environment and special equipments that can work in the environment. For example, EB lithography [1-5], the next generation IC lithography, requires high vacuum around 10^{-3}–10^{-4} Pa without any electromagnetic field, since trajectory of electron beam is easily affected by external electromagnetic field. EB lithography also requires positioning stages that can work in the high vacuum without leaking any electromagnetic field [4,5].

Conventional electromagnetic motors do not seem suitable for those applications. They utilize strong magnetic field from their magnets and armatures for their thrust / torque generations. The strong magnetic field is very hard to be shielded and results in large leakage of electromagnetic field, which is most disliked in the above applications. If we still try to introduce conventional electromagnetic motor into those applications in despite of the problem, we have to put huge magnetic shield onto the motor to shut out the leakage. This makes the whole system bigger, and waste valuable space of vacuum chamber.

On the other hand, the authors have developed high-power electrostatic linear motor [6-8]. The linear motor, called *Dual Excitation Multiphase Electrostatic Drive* (DEMED), is a direct-drive linear motor driven by high voltage such as several kilo volts. Differently from micro electrostatic motors [9-12], DEMED is sized several centimeters and has the capability of far larger thrust force generation from several tens to several hundreds of newtons, depending on motor size. Since DEMED is driven by relatively high voltage, drive current is correspondingly low, if compared with electromagnetic motors of the same output power. Due to this characteristic, high voltage and low current, DEMED generates stronger electrostatic field but weaker magnetic field. Since electrostatic field is easily shielded even by thin conductive plate, the leakage of electrostatic field from DEMED is easily extinguished. That means DEMED leaks far weaker electromagnetic field than conventional electromagnetic motors.

Due to the mentioned characteristics such as large thrust force and little electromagnetic field leakage, DEMED seems suitable for the above vacuum applications. To evaluate the performance of DEMED in vacuum, we had fabricated linear stage using DEMED, which linear guide is specially treated for vacuum use [13]. In this study, we investigate its thrust force characteristics in high vacuum of 10^{-3}Pa. Then, we have developed built-in position sensor. The sensor detects the slider position by using motor's electrodes. Since this sensor system requires no additional electrodes, it contributes to reducing size of whole motor system. Finally, positioning control has been carried out using developed position sensor.

II. Overview of *Dual Excitation Multiphase Electrostatic Drive*

A basic structure of DEMED is shown in fig. 1. Basic DEMED consists of a pair of flat films, slider and stator, respectively. Both films are made using flexible printed circuit board (FPC) technology. They contain 200μm pitch three-phase electrodes. As shown in fig. 1, slider electrodes are serrated in shape so as to reduce thrust force ripple, whereas stator electrodes are straight [7].

In previous research on DEMED, DEMED has been evaluated in non-vacuum condition (in the atmosphere) [6-8]. In the case, two films are just stacked without using linear guide. Glass beads with diameter of 20μm are inserted

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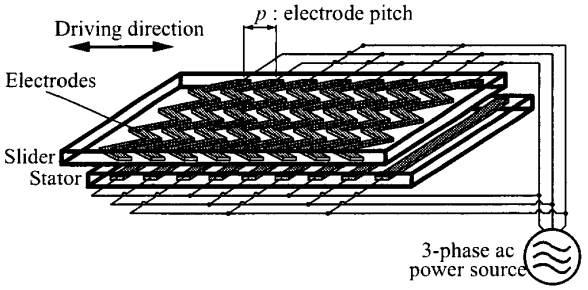


Fig. 1 Basic structure of DEMED

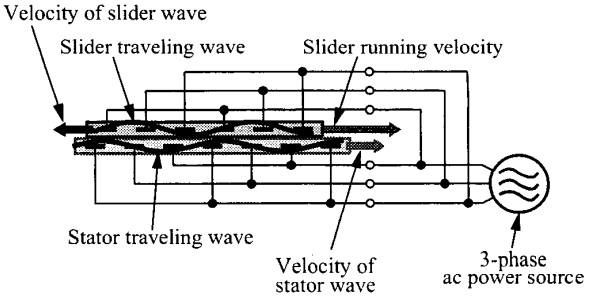


Fig. 2 Driving principle of DEMED

between two films to reduce the friction caused as slider moves. Besides, they are immersed in dielectric liquid to prevent dielectric breakdown of the air, since electrical breakdown field strength is very low for the atmospheric air. Since the purpose of using dielectric liquid is to prevent electrical breakdown of the atmospheric air, those liquid is not required in vacuum. We can directly apply high voltage in vacuum condition without causing breakdown, as *Paschen's law* on electrical breakdown states.

Fig. 2 shows driving principle. DEMED is a synchronous electrostatic drive that is driven by three-phase ac voltage. By applying three-phase ac voltage onto both three-phase of stator and slider electrodes, sinusoidal potential distributions are excited on the two films. Since connections of three-phase ac are different in stator and slider, two potential distributions run toward opposite directions to each other at a speed of $\pm u_d$. There is electrostatic interaction between two potential distributions, and the interaction occurs so as to keep stable spatial phase difference between the two distributions. Resultantly, in synchronous state, slider runs at a speed of $2u_d$ so as to keep the spatial phase difference almost constant. Namely, the slider speed u is expressed as:

$$u = 2\,u_d = 6\,pf, \tag{1}$$

where p is electrode pitch and f is frequency of applied ac voltage.

Thrust force generation was analyzed and tested in past research [8,13]. According to the analysis, thrust force f_x can be expressed as

$$f_x = \kappa_t v_0^2 \sin\left(2\phi + \frac{2\pi}{3p}x + \frac{\pi}{3}\right), \tag{2}$$

for the applied voltage

$$v = v_0\,[\sin(\phi), \sin(\phi - 2\pi/3), \sin(\phi + 2\pi/3)]. \tag{3}$$

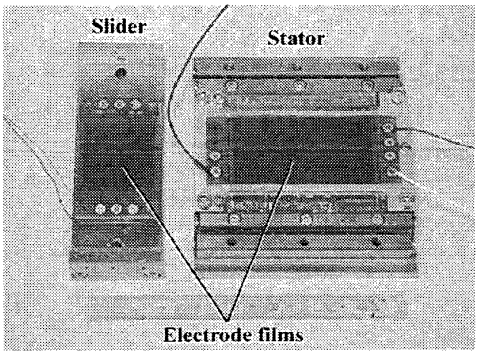
Constant κ_t in (2) is thrust force coefficient that is determined by motor's geometrical factors. Variable x denotes slider position.

III. Prototype Linear Stage for Use in Vacuum

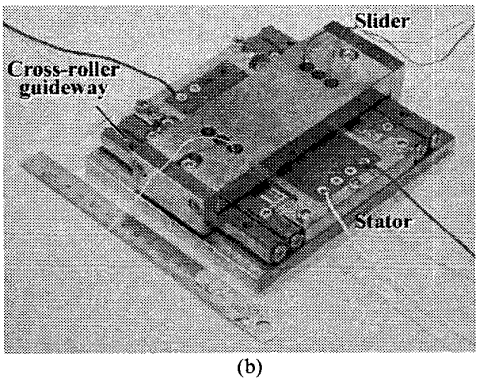
In the past research on DEMED, we have adopted simple structure as schematically shown in fig. 1, where two plates (slider and stator) are just stacked without using linear guide, and slider slides across the stator with causing friction. However, such simple structure is not suitable for use in vacuum since friction is known to increase in vacuum. In this work, we built-up a linear stage specially fabricated for vacuum use.

A. Structure

Fig. 3(a) and 3(b) show the developed prototype linear motor stage. Fig. 3(a) shows the surfaces of stator and slider, and 3(b) shows the fabricated setup. Stator and slider are made of stainless, on which films that contain three-phase electrodes of 200μm pitch are glued with epoxy resin. Slider is guided by linear cross-roller guide which rollers are



(a)



(b)

Fig. 3 Prototype linear stage

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coated with fluorine polymer to prevent adhesion in vacuum environment. The slider weighs 1.08kg.

B. Thrust force measurement in the atmosphere

Before bringing this linear stage into vacuum condition, we measured thrust force characteristics in the atmosphere. The measurement was done by weight-lifting experiments. In the experiments, thrust force was simply determined by finding maximum weight that the stage was able to lift up.

Theoretically, thrust force of DEMED is proportional to square of the applied voltage amplitude. Therefore, higher voltage results in larger thrust force. However, in the atmosphere, as mentioned above, we can not apply enough high voltage to obtain measurable thrust force, because electric breakdown field strength of the atmosphere is far lower than that of vacuum. For that reason, the measurement experiments were done by filling up the gap between slider and stator with dielectric liquid called *Fluorinert* (FC-77, 3M).

The results of the measurements are plotted in fig. 4. As theoretically predicted, thrust force was proportional to square of voltage.

C. Performances in vacuum with open loop control system

We brought the prototype linear motor stage into vacuum chamber, and investigated open loop characteristics at the pressure of 10^{-3}Pa.

First, we operated the motor in an open loop control with three-phase ac of ±1Hz. The synchronous speed is calculated as ±1.2mm/s. The frequency was alternated +1Hz and −1Hz every three seconds, to cause reciprocating motion of the slider. The amplitude of the applied voltage v_0 was 1.5kV. The measured motion of the slider is plotted in fig. 5. The motion was measured using optical displacement sensor ZIMMER model 200, which can observe the motion of the target optically through the view port from outside the chamber. From fig. 5, it is verified that this motor can work properly in vacuum condition. Besides, we succeeded to drive it continuously for 15 hours without noticeable temperature raise.

Next, we estimated the thrust force in vacuum condition. Since it is considerably a difficult task to measure thrust force directly in vacuum, we estimated thrust from pull-in characteristics. Since DEMED is a synchronous drive, if we apply a driving voltage of a certain constant frequency to a

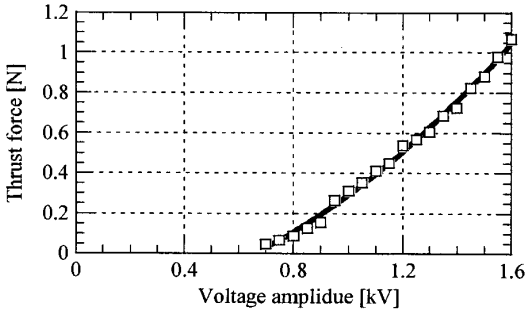


Fig. 4 Thrust force measured in dielectric liquid

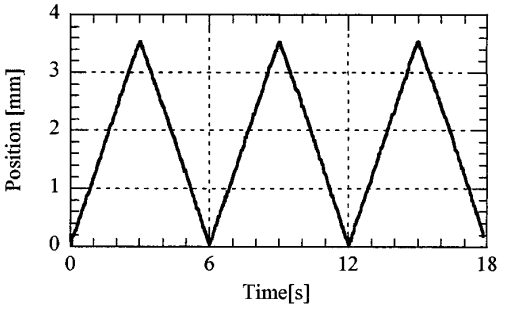


Fig. 5 Slider displacement in open loop operation at 10^{-3}Pa

motor in a stationary state, it will react by one of two ways: one reaction is being unable to enter synchronous state and not moving; the other is entering synchronous state and starting to move at the synchronous speed. The threshold frequency for these two reactions is determined by thrust force; larger thrust force results in higher threshold frequency.

At the threshold frequency f_{max}, where the motor can barely enter the synchronous state, the slider must obtain impulse to reach the momentum that is equivalent to the synchronous speed within a certain period that is proportional to the cycle of applied frequency. Since the synchronous speed is $6pf_{max}$, the momentum m_s that the slider obtains is expressed as

$$m_s = k_m\, m\, f_{max}\,, \tag{4}$$

with a proportionality factor k_m. Here, m denotes the mass of the slider. On the other hand, the impulse i_m given by thrust force for the certain period is

$$i_m = k_i \frac{f_x}{f_{\max}}, \tag{5}$$

where k_i is a proportionality factor. At f_{max}, the impulse i_m must be equal to the momentum m_s, so we obtain the next condition from (4) and (5);

$$f_x = k\, m f_{max}^{\,2}\,, \tag{6}$$

where $k = k_m / k_i$. We can not theoretically determine the factor k, however, those factors are constant for both in the atmosphere and in vacuum. Therefore, we can estimate thrust force in vacuum from that in the atmosphere by comparing mf_{max}^2 for both conditions.

First, we measured mf_{max}^2 in the atmosphere as shown in fig. 6. By comparing the result with fig. 4, we found $k = 4.0\text{x}10^{-3}$ [$\text{N}\cdot\text{kg}^{-1}\cdot\text{s}^2$]. The result in fig. 4 is re-plotted in fig. 6 by using the obtained value of k. Then, we measured mf_{max}^2 in vacuum condition as also shown in fig. 6. Using the k value, thrust force in vacuum condition was estimated. It was found that the stage in vacuum performed almost the same thrust force characteristics as in the atmosphere. The maximum thrust force in vacuum was estimated as 1.2N at applied voltage of $1.6\text{kV}_{0\text{-p}}$.

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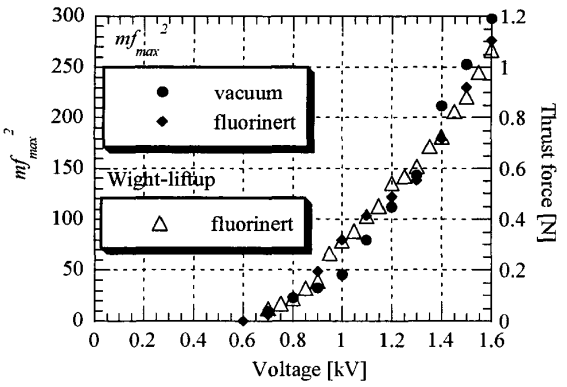


Fig. 6 Estimated thrust force

IV. Built-in Position Sensor

To realize closed loop control system, we need position sensor. However, there are very few position sensors available in vacuum condition. One reasonable solution is to use laser displacement sensor by transmitting laser from outside chamber through view port, or to use optical sensor such as ZIMMER model 200, which is introduced in the previous section. However, those sensors usually require large systems. In this study, in consideration of simplicity of whole system, we have developed special built-in position sensor. In previous research, we have developed capacitance linear scale sensor system, whose electrodes are easily integrated into the motor's films [8]. However, it requires additional area for sensor electrodes on the films, and makes the motor size larger. In this paper, we propose a new capacitance linear scale sensor system that can work using motor's driving electrodes.

Fig. 7 shows a block diagram of the sensor system. Sensor signals of 253 kHz three-phase ac are superposed onto the driving signal lines using transformers and picked out using another set of transformers. Since capacitances composed by the motor electrodes vary with slider position, currents of the sensor signals also vary with it. Resultantly, computing sensor signals picked out through transformers gives slider displacement.

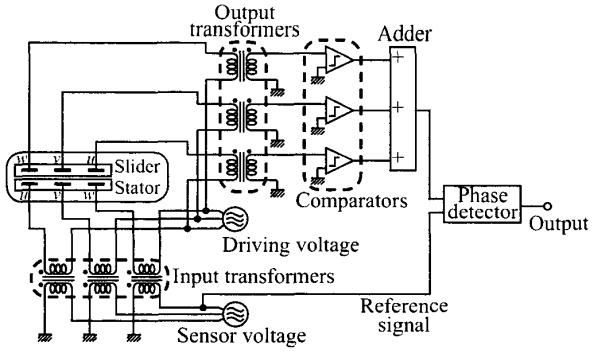


Fig. 7 Sensor circuit diagram

The sensor principle is analyzed as follows. A simplified equivalent circuit for sensor signals is expressed as shown in fig. 8. Let $\boldsymbol{i} = [i_u, i_v, i_w]^t$ be the currents through three input coils, and $\boldsymbol{v} = [v_u, v_v, v_w]^t$ be the voltage differences between stator electrode and slider electrode of the same phase. The authors already analyzed capacitance of the motor electrodes in the previous research [8]. According to the analysis, relationship between $\boldsymbol{i}$ and $\boldsymbol{v}$ is given as $\boldsymbol{i} = \boldsymbol{C}(\mathrm{d}\boldsymbol{v}/\mathrm{d}t)$, where

$$\boldsymbol{C} = \begin{bmatrix} C_0 + C_2\cos\left(\frac{2\pi}{3p}x - \frac{2\pi}{3}\right) & C_1 + C_2\cos\left(\frac{2\pi}{3p}x + \frac{2\pi}{3}\right) & C_1 + C_2\cos\left(\frac{2\pi}{3p}x\right) \\ C_1 + C_2\cos\left(\frac{2\pi}{3p}x + \frac{2\pi}{3}\right) & C_0 + C_2\cos\left(\frac{2\pi}{3p}x\right) & C_1 + C_2\cos\left(\frac{2\pi}{3p}x - \frac{2\pi}{3}\right) \\ C_1 + C_2\cos\left(\frac{2\pi}{3p}x\right) & C_1 + C_2\cos\left(\frac{2\pi}{3p}x - \frac{2\pi}{3}\right) & C_0 + C_2\cos\left(\frac{2\pi}{3p}x + \frac{2\pi}{3}\right) \end{bmatrix}. \quad (7)$$

In the matrix, C_0, C_1, and C_2 are constants defined by form factor of the electrodes. These equations give currents as

$$\boldsymbol{i} = \begin{bmatrix} \alpha\cos(\omega t) + \beta\cos\left(\omega t - \frac{2\pi}{3p}x + \frac{2\pi}{3}\right) \\ \alpha\cos(\omega t - \frac{2\pi}{3}) + \beta\cos\left(\omega t - \frac{2\pi}{3p}x - \frac{2\pi}{3}\right) \\ \alpha\cos(\omega t + \frac{2\pi}{3}) + \beta\cos\left(\omega t - \frac{2\pi}{3p}x\right) \end{bmatrix}, \quad (8)$$

where ω is frequency of sensor signals, and α and β are constants defined by C_0, C_1, C_2, ω, amplitude of sensor voltage, and inductance of transformers L.

These currents are converted to voltages by output coils as $\boldsymbol{V_{out}} = [V_{ou}, V_{ov}, V_{ow}]^t = (\mathrm{d}\boldsymbol{i}/\mathrm{d}t)L$. Apparently, the current $\boldsymbol{i}$ or the output voltage $\boldsymbol{V_{out}}$ contains an information about position x in their phase. Therefore, we can calculate x by processing $\boldsymbol{i}$ or $\boldsymbol{V_{out}}$. Our approach is as follows: First we

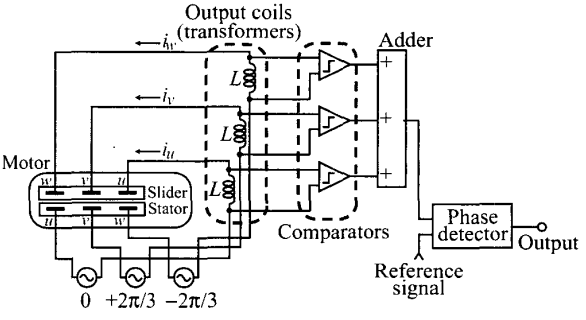


Fig.8 Equivalent circuit for sensor signals

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feed V_{ou}, V_{ov}, and V_{ow} to voltage comparators (against zero volt) to make square waves, then added them into one, which is fed into lock-in amplifier. Lock-in amplifier detects the phase difference between the added signal and sensor input voltage. The output of lock-in amplifier is plotted in fig. 9. The output repeats every 600μm. We found 50μm linearity error at the maximum.

V. Positioning Servo Control

Finally, positioning servo control in vacuum condition was carried out using developed built-in position sensor.

A. Control strategy

The thrust force of DEMED depends on three variables, voltage amplitude v_0, phase ϕ, and slider position x as expressed in (2). In a servo control system, those three variables should be controlled so as to generate desired thrust force. However, it is rather complicated to control three variables independently, we had proposed to correlate phase ϕ with slider position x [8]. Since v_0 is limited by electrical breakdown voltage of the electrode film (around 2–3kV), the thrust force is also limited by it. To maximize the thrust force limit, we had proposed to determine the phase ϕ as

$$\phi = -\frac{\pi}{3p}x - \frac{\pi}{6} \pm \frac{\pi}{4} \quad . \tag{9}$$

With (9), we can maximize the thrust force at each v_0, and then, thrust force can be re-written as

$$f_x = \pm\kappa_t v_0^2 . \tag{10}$$

We established positioning servo controller based on the above strategy. Fig. 10 shows the block diagram of the controller. The main part of the system was implemented on DSP system (d-space, DS1104). In the *3-phase signal generator*, which is also implemented on DSP, three-phase signal $\boldsymbol{v} = v_0$ [sin(ϕ), sin(ϕ–2π/3), sin(ϕ+2π/3)] is generated as:

$$v_0 = \sqrt{\frac{\left|f_x^*\right|}{\kappa_t}} \quad , \tag{11}$$

$$\phi = \begin{cases} -\dfrac{\pi}{3p}x + \dfrac{\pi}{12} & (\text{in case } f_x^* \geq 0) \\ -\dfrac{\pi}{3p}x - \dfrac{5\pi}{12} & (\text{in case } f_x^* < 0) \end{cases} , \tag{12}$$

based on (9) and (10). The generated signals are output from DSP, then fed into high-voltage amplifiers (AS-123, NF Corporation) to be amplified by 1000 before being applied to the motor.

B. Step response

The response against step input of 10mm is shown in fig. 11. The vertical axis, labeled as "Position", is the output of the built-in position sensor. Fine step response was successfully demonstrated, however, steady-state error of around 20μm is observed. This is probably caused by friction of linear cross-roller guide, since friction is known to increase in vacuum. In our future works, friction compensation will be implemented into controller to reduce steady-state error.

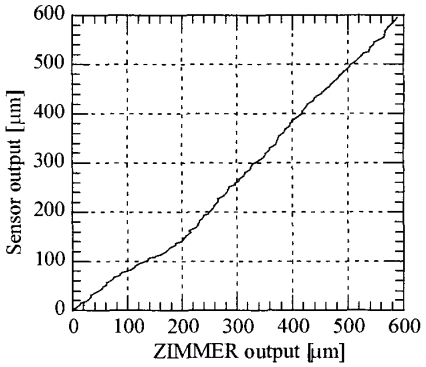


Fig. 9 Built-in sensor output compared with ZIMMER

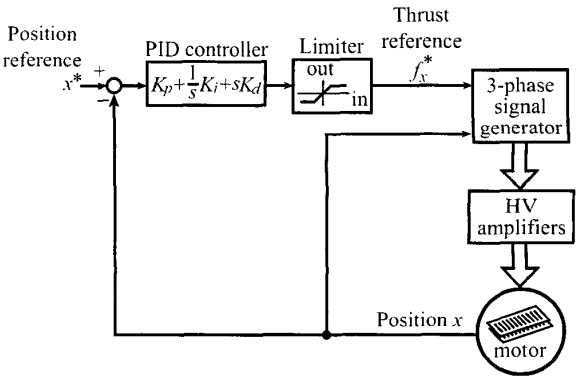


Fig. 10 Positioning servo controller

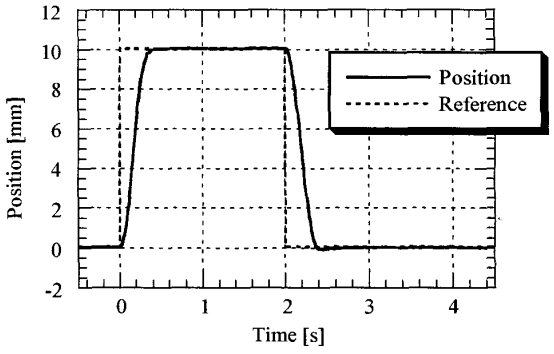


Fig. 11 Step response of positioning control

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C. Evaluation by external position sensor

In the experiment of the previous section, the built-in position sensor was used for stage control, as well as for the evaluation. However, the built-in sensor has linearity error of 50μm at the maximum as shown in fig. 9, and therefore, the evaluation using the sensor is not perfectly correct. To evaluate the control performance more precisely, we observed the real stage displacement by the Optical displacement sensor ZIMMER model 200.

Fig. 12 shows the result of positioning control. In the control experiments, reference position value changed by 1mm at every 1 second. Positioning control was done only with the built-in position sensor, and ZIMMER 200 was used only for evaluation. Fig. 13 shows the distribution of positioning error evaluated by ZIMMER 200. Steady-state positioning error for each step was found as the average error value during a period from 0.5 second after each step input until the next step input. All positioning errors were distributed within ±100μm.

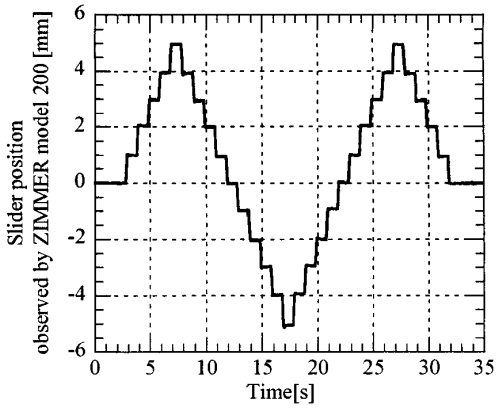


Fig. 12 Positioning control at 1mm steps

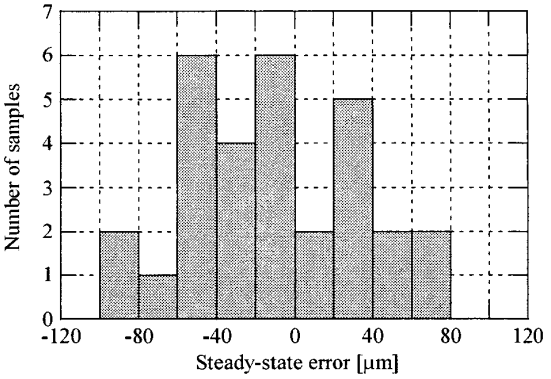


Fig. 13 Distribution of position errors

VI. Conclusions

In this paper, we reported development of electrostatic linear motor stage with built-in position sensor that was developed for use in vacuum condition,.

First, we fabricated linear motor stage, which was brought into high vacuum of around 10^{-3}Pa and tested in open loop system. It performed fine operation, and its maximum thrust force was estimated as 1.2N.

Secondly, we developed built-in position sensor, which utilizes the same electrodes as motor driving. This feature contributes to compactness of whole motor system.

Finally, positioning servo control in vacuum condition was demonstrated using built-in position sensor. The control system demonstrated fine response, however, steady-state error due to friction and sensor error was observed. In our future work, it will be improved by implementing friction compensation into controller and also by improving the sensor characteristics.

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